

TrustGuard: A Privacy-First Continuous Identity Trust Framework for Secure Digital Banking

Project Overview

TrustGuard is a privacy-first continuous identity trust platform that protects digital banking systems against account takeover attacks through real-time risk assessment and adaptive authentication. By continuously evaluating contextual and behavioral signals, the platform enables secure, friction-optimized access while minimizing unnecessary authentication challenges.

The system focuses on detecting suspicious login attempts, new device access, abnormal user behavior, and fraudulent account recovery requests while minimizing authentication friction for genuine users.

Problem Statement

Digital banking platforms are increasingly vulnerable to account takeover attacks, where attackers gain access using stolen credentials, compromised devices, or social engineering techniques.

Traditional authentication mechanisms verify users only at login and cannot continuously determine whether the person accessing the account remains trustworthy throughout the session. At the same time, excessive security checks negatively impact user experience.

There is a need for a risk-based framework that can continuously assess identity trust and trigger additional verification only when risk levels become elevated.

Proposed Solution

TrustGuard continuously monitors user activities and calculates a dynamic Trust Score based on factors such as:

- Device information
- Login patterns
- Location changes
- Failed login attempts
- Account recovery requests

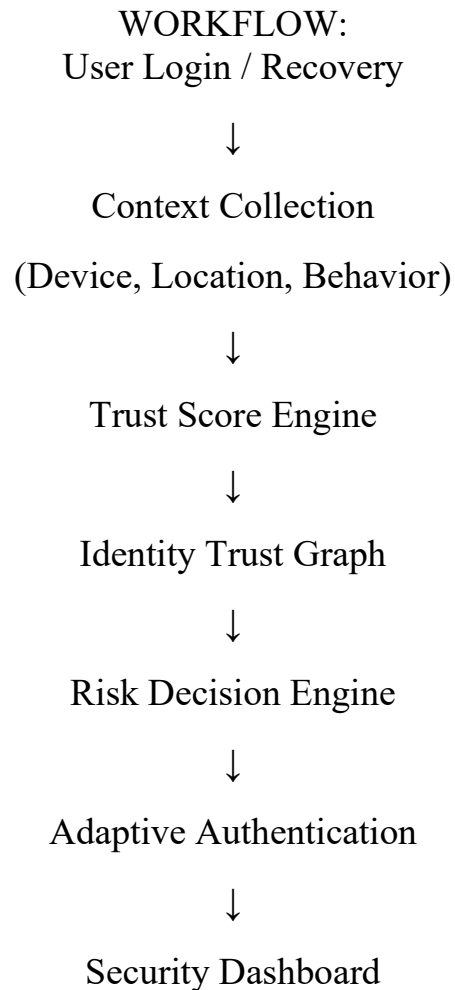
- Behavioral anomalies

The Trust Score determines the level of authentication required.

For example:

- Low-risk activity → Access granted
- Medium-risk activity → OTP verification
- High-risk activity → Additional verification
- Critical-risk activity → Access blocked and alert generated

This approach improves security without creating unnecessary inconvenience for legitimate users.



Key Innovation: Identity Trust Graph

The primary innovation of TrustGuard is the Identity Trust Graph.

The graph establishes relationships between:

- Users
- Devices
- IP Addresses
- Accounts
- Login Events
- Recovery Requests

By analyzing these relationships, the system can identify suspicious patterns such as:

- Multiple accounts using the same device
- Repeated login attempts from unknown devices
- Suspicious account recovery activities
- Shared high-risk network sources

This allows earlier detection of potential account takeover attempts.

Risks Addressed

Primary Risk:

- Account Takeover (ATO)

Supporting Risks:

- New Device Risk
- Behavioral Anomalies
- Suspicious Account Recovery

Proposed System Modules

Customer Portal

- User Login
- Authentication Workflow
- Trust Score Display

Trust Score Engine

- Risk Calculation
- Risk Classification
- Adaptive Authentication

Identity Trust Graph

- Relationship Mapping
- Suspicious Connection Detection

Security Dashboard

- Risk Monitoring
- Alert Generation
- Trust Score Tracking

Expected Outcomes

- Reduction in account takeover incidents
- Early detection of suspicious login activities
- Improved account recovery security
- Reduced authentication friction for genuine users
- Better visibility into identity-related risks
- Scalable deployment across digital banking channels

Proposed Prototype

The prototype will demonstrate:

- Customer Login Portal
- Continuous Trust Score Calculation
- Identity Trust Graph Visualization
- Adaptive Authentication
- Security Dashboard with Real-Time Alerts

Proposed Technology Stack

- ☐ **Frontend:** Next.js, Tailwind CSS
- ☐ **Backend:** FastAPI (Python)
- ☐ **Database:** PostgreSQL
- ☐ **Graph Analytics:** NetworkX / Neo4j
- ☐ **Authentication:** JWT
- ☐ **Deployment:** Docker

Conclusion

TrustGuard introduces a continuous identity trust approach that moves beyond traditional authentication by combining contextual risk analysis, relationship-based identity mapping, and adaptive authentication. The proposed solution aims to reduce account takeover attacks while maintaining a secure, privacy-first, and seamless digital banking experience.